

Qualification Pack



Aerospace Structural Fitter

QP Code: AAS/Q1602

Version: 3.0

NSQF Level: 3

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AAS/Q1602: Aerospace Structural Fitter

Brief Job Description

Aerospace Structural Fitter is responsible for carrying out structural assembly of aircrafts/ helicopters and perform activities like drilling, reaming, riveting of structures, applying sealing compound etc.

Personal Attributes

The individual should have basic communication ability and should be responsible in order to maintain procedures. He should have the ability of operation monitoring i.e., observing gauges, dials etc., ability to quickly move hand to grasp and assemble objects (dexterity), reading, writing and communication skills and sensitivity towards safety for self and equipment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AAS/N1616: Understand the Assembly Drawings.](#)
2. [AAS/N1617: Understanding the Different Drilling Operations.](#)
3. [AAS/N1618: Understanding of Different types of Riveting Operations](#)
4. [AAS/N1619: Understand the Reaming Operation](#)
5. [AAS/N1620: Understand the Canopy Matching](#)
6. [AAS/N1621: Perform RMA installation of the Aircraft](#)
7. [AAS/N1622: Mixing of Sealants and Application of Sealants](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Aerospace and Aviation
Sub-Sector	Manufacturing & Assembly
Occupation	Aircraft Assembly, Testing and Overhauling
Country	India

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NSQF Level	3
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8211.1400
Minimum Educational Qualification & Experience	10th grade pass (Or Equivalent) OR Previous relevant Qualification of NSQF Level (2) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-03-AA-04189-2025-V2-AASSC
NQR Version	3

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AAS/N1616: Understand the Assembly Drawings.

Description

This module focuses on developing the ability to read and interpret assembly drawings used in aerospace manufacturing. It trains individuals to understand various views, sections, symbols, scales, and part identification in coordination with process sheets and technology, ensuring accurate interpretation and execution of technical drawings.

Scope

The scope covers the following :

- Reading Assembly Drawing,
- know-how to make use of the drawing signs, drawing scale, drawing number,
- Description of drawing in coordination of Process sheet & Technology.

Elements and Performance Criteria

Basic Knowledge of the Drawing

To be competent, the user/individual on the job must be able to:

- PC1.** Able to read the drawing
- PC2.** Know the views of drawing i.e. plan, isometric etc.
- PC3.** Able to identify sections of drawing.
- PC4.** Know the types of lines used to differentiate between the assembly part positions.
- PC5.** Know the signs of drawing like finishing triangles, hole size, thickness, length, width etc.
- PC6.** Know to identify the scale of drawing
- PC7.** Able to correlate the drawing and technology
- PC8.** Know to identify the position numbers of the part and able to identify the part number in material schedule (MS)
- PC9.** Know how to see the section of drawing in different zones
- PC10.** Mark the center of the hole with a punch.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the role of technical drawings in aerospace manufacturing
- KU2.** Recognize types of views (plan, front, side, isometric, sectional).
- KU3.** Interpret line types for assembly positions
- KU4.** Identify symbols for hole size, thickness, length, and width.
- KU5.** Read and apply drawing scales.
- KU6.** Understand drawing numbers, title blocks, and revisions.
- KU7.** Correlate assembly drawings with process sheets and technology.

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- KU8.** Identify part and position numbers in the material schedule.
- KU9.** View and locate sections within different zones of a drawing.
- KU10.** Mark hole centers using a punch tool.
- KU11.** Visualize 2D drawings for 3D assembly.
- KU12.** Coordinate with planners using drawings for proper fitment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret engineering drawings accurately.
- GS2.** Communicate effectively with team members and supervisors.
- GS3.** Perform basic mathematical calculations related to measurements and tolerances.
- GS4.** Use measuring instruments (e.g., scale, callipers, micrometres) accurately.
- GS5.** Visualize and interpret 2D and 3D views, including isometric and sectional drawings.
- GS6.** Identify drawing symbols, line types, and notations correctly.
- GS7.** Select appropriate tools based on drawing and operation requirements.
- GS8.** Handle hand tools and basic power tools safely and efficiently.
- GS9.** Pay close attention to detail while marking, measuring, and assembling parts.
- GS10.** Follow process flow and routing as per organizational procedures.
- GS11.** Maintain proper documentation and update records as needed.
- GS12.** Operate basic computer applications for viewing or accessing digital drawings.
- GS13.** Collaborate with other team members in a coordinated and efficient manner.
- GS14.** Follow safety guidelines and company-specific compliance procedures.
- GS15.** Solve minor fitting and alignment issues using logical problem-solving approaches.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic Knowledge of the Drawing</i>	40	61	-	-
PC1. Able to read the drawing	4	6	-	-
PC2. Know the views of drawing i.e. plan, isometric etc.	4	6	-	-
PC3. Able to identify sections of drawing.	4	6	-	-
PC4. Know the types of lines used to differentiate between the assembly part positions.	4	6	-	-
PC5. Know the signs of drawing like finishing triangles, hole size, thickness, length, width etc.	4	6	-	-
PC6. Know to identify the scale of drawing	4	6	-	-
PC7. Able to correlate the drawing and technology	4	6	-	-
PC8. Know to identify the position numbers of the part and able to identify the part number in material schedule (MS)	4	6	-	-
PC9. Know how to see the section of drawing in different zones	4	6	-	-
PC10. Mark the center of the hole with a punch.	4	7	-	-
NOS Total	40	61	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1616
NOS Name	Understand the Assembly Drawings.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AAS/N1617: Understanding the Different Drilling Operations.

Description

This module introduces key drilling operations such as drilling, reaming, tapping, and countersinking. It focuses on their application, tool selection, and execution as per technical specifications in aerospace assembly.

Scope

The scope covers the following :

- Responsible for conducting drilling activities like: Drilling, Counter sinking, Counter boring, Spot facing, Reaming & Taping.
- Responsible for measuring dimensions and inspecting work pieces.
- Carry out machining activities to make Aerospace Assembly.
- Interact with other technicians, supervisors, inspectors, process planner, design, tool planner etc.

Elements and Performance Criteria

Perform different drilling operation

To be competent, the user/individual on the job must be able to:

- PC1.** Read the process information sheet & drawing and operations.
- PC2.** Check and inspect the parts supplied for physical / dimensional variation.
- PC3.** Check the documentation of all parts and its identification with respect to documents.
- PC4.** Marking on the part according to drawing / technology/ reference tool.
- PC5.** Check whether the component having proper edge distance and other dimensions prior to carrying out drilling operation.
- PC6.** Select the proper tool as mentioned in the technology.
- PC7.** Check the machinability of tool i.e. like sharpness, size etc.
- PC8.** Decide the number of stages in which the hole is to be enlarges as per size and its steps.
- PC9.** Select the tools as per number of stages in which drilling is to be carried out.
- PC10.** Mark the centre of the hole by punch or perform drilling on pilot holes.
- PC11.** After each stage check the dimension of the operation performed by tool mentioned in the technology.
- PC12.** Deburr the hole by tool as mentioned in technology.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand various drilling operations: drilling, reaming, tapping, countersinking.
- KU2.** Know how to read process sheets and technical drawings.
- KU3.** Identify and inspect parts for physical or dimensional issues.

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- KU4.** Mark components accurately based on drawings.
- KU5.** Check edge distances and required dimensions before drilling.
- KU6.** Select appropriate tools as per the operation.
- KU7.** Verify tool sharpness, size, and machinability.
- KU8.** Plan multi-stage drilling processes as per hole size.
- KU9.** Perform pilot hole drilling and center marking with a punch.
- KU10.** Measure hole dimensions after each stage.
- KU11.** Deburr holes using specified tools.
- KU12.** Coordinate with technical teams for accurate execution.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret process sheets and technical drawings accurately.
- GS2.** Communicate effectively with team members, supervisors, and other departments.
- GS3.** Use measuring tools to verify dimensions and tolerances.
- GS4.** Select and handle tools correctly for different drilling operations.
- GS5.** Pay attention to detail during marking, drilling, and inspection.
- GS6.** Plan and execute multi-stage drilling processes as per specifications.
- GS7.** Maintain safety and follow standard operating procedures.
- GS8.** Record and interpret job-related documentation and reports.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform different drilling operation</i>	40	60	-	-
PC1. Read the process information sheet & drawing and operations.	3	5	-	-
PC2. Check and inspect the parts supplied for physical / dimensional variation.	3	5	-	-
PC3. Check the documentation of all parts and its identification with respect to documents.	3	5	-	-
PC4. Marking on the part according to drawing / technology/ reference tool.	3	5	-	-
PC5. Check whether the component having proper edge distance and other dimensions prior to carrying out drilling operation.	3	5	-	-
PC6. Select the proper tool as mentioned in the technology.	3	5	-	-
PC7. Check the machinability of tool i.e. like sharpness, size etc.	4	5	-	-
PC8. Decide the number of stages in which the hole is to be enlarges as per size and its steps.	3	5	-	-
PC9. Select the tools as per number of stages in which drilling is to be carried out.	4	5	-	-
PC10. Mark the centre of the hole by punch or perform drilling on pilot holes.	4	5	-	-
PC11. After each stage check the dimension of the operation performed by tool mentioned in the technology.	3	5	-	-
PC12. Deburr the hole by tool as mentioned in technology.	4	5	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1617
NOS Name	Understanding the Different Drilling Operations.
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AAS/N1618: Understanding of Different types of Riveting Operations

Description

This module focuses on performing riveting operations on aerospace panels. It includes reading technical drawings, selecting appropriate tools and rivets, cleaning components, executing precise riveting, and inspecting completed work. The technician is also responsible for documenting measurements and coordinating with other departments to ensure quality and compliance with specifications.

Scope

The scope covers the following :

- Responsible for conducting riveting activities on panels.
- Responsible for measuring dimensions and inspecting work pieces.
- Carry out machining activities to make Aerospace Assembly.
- Interact with other technicians, supervisors, inspectors, process planner, design, tool planner etc.

Elements and Performance Criteria

Perform different riveting operations

To be competent, the user/individual on the job must be able to:

- PC1.** Read the process information sheet & drawing and operations.
- PC2.** Check and inspect the parts supplied for physical / dimensional variation.
- PC3.** Check the documentation of all parts and its identification with respect to documents.
- PC4.** Marking on the part according to drawing / technology/ reference tool.
- PC5.** Check whether the component having proper cleaning for riveting operation.
- PC6.** Select the proper tool as mentioned in the technology.
- PC7.** Select the rivet as mentioned in the technology.
- PC8.** Clean all the parts and rivets with naphtha for riveting.
- PC9.** Now parts are tech. bolted, and rivets are inserted on holes with primer.
- PC10.** If sealing compound is called on technology, then apply sealing instead of primer.
- PC11.** Do the riveting with the rivet gun on one side and bucking bar on another side.
- PC12.** Measure the bucking head and bucking length by Rivet Gauge.
- PC13.** If riveting is improper then rivets it and install new rivet and do the riveting.
- PC14.** Document the dimension.
- PC15.** Apply the non-corrosive chemical (Primer) as per the technology.
- PC16.** Offer to supervisor and inspect.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of reading and interpreting process sheets, technical drawings, and operations.

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- KU2.** Understanding of physical and dimensional inspection techniques for parts and components.
- KU3.** Knowledge of documentation procedures and part identification standards.
- KU4.** Understanding of marking methods based on drawings, technology, or reference tools.
- KU5.** Knowledge of proper cleaning methods for parts before riveting operations.
- KU6.** Understanding of selecting the appropriate tools and rivets as per the technology and process requirements.
- KU7.** Knowledge of cleaning parts and rivets with naphtha prior to riveting.
- KU8.** Understanding the process of tech bolting parts and inserting rivets with primer or sealing compound.
- KU9.** Knowledge of using rivet guns and bucking bars for proper riveting.
- KU10.** Understanding how to measure bucking head and length using a Rivet Gauge.
- KU11.** Knowledge of how to address improper riveting by replacing rivets and redoing the process.
- KU12.** Knowledge of documenting dimensions and keeping track of quality control.
- KU13.** Understanding the application of non-corrosive chemicals (e.g., primer) as specified in the technology.
- KU14.** Knowledge of offering parts for inspection and coordinating with supervisors for final checks.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret process sheets, technical drawings, and work instructions accurately.
- GS2.** Use measuring tools (e.g., rivet gauge, scale) to verify dimensions and tolerances.
- GS3.** Select and handle tools and rivets appropriately for the operation.
- GS4.** Follow safety procedures while working with chemicals, tools, and machinery.
- GS5.** Pay attention to detail during marking, cleaning, riveting, and inspection.
- GS6.** Identify and correct improper riveting using approved methods.
- GS7.** Maintain proper documentation of dimensions and work status.
- GS8.** Coordinate with other departments and team members to ensure smooth workflow.
- GS9.** Follow organizational protocols and quality standards throughout the operation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform different riveting operations</i>	40	60	-	-
PC1. Read the process information sheet & drawing and operations.	3	3	-	-
PC2. Check and inspect the parts supplied for physical / dimensional variation.	3	3	-	-
PC3. Check the documentation of all parts and its identification with respect to documents.	3	3	-	-
PC4. Marking on the part according to drawing / technology/ reference tool.	3	3	-	-
PC5. Check whether the component having proper cleaning for riveting operation.	3	4	-	-
PC6. Select the proper tool as mentioned in the technology.	3	4	-	-
PC7. Select the rivet as mentioned in the technology.	3	4	-	-
PC8. Clean all the parts and rivets with naphtha for riveting.	3	4	-	-
PC9. Now parts are tech. bolted, and rivets are inserted on holes with primer.	2	4	-	-
PC10. If sealing compound is called on technology, then apply sealing instead of primer.	2	4	-	-
PC11. Do the riveting with the rivet gun on one side and bucking bar on another side.	2	4	-	-
PC12. Measure the bucking head and bucking length by Rivet Gauge.	2	4	-	-
PC13. If riveting is improper then rivets it and install new rivet and do the riveting.	2	4	-	-
PC14. Document the dimension.	2	4	-	-
PC15. Apply the non-corrosive chemical (Primer) as per the technology.	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Offer to supervisor and inspect.	2	4	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1618
NOS Name	Understanding of Different types of Riveting Operations
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AAS/N1619: Understand the Reaming Operation

Description

This module covers the execution of reaming operations in aerospace assembly, focusing on achieving precise hole dimensions and surface finish. It includes selecting appropriate tools and coolants, setting correct machining parameters, and inspecting workpieces per technical specifications. The role also involves thorough documentation and maintaining cleanliness and corrosion prevention of tools and components.

Scope

The scope covers the following :

- Responsible for conducting machining activities like reaming operations.
- Responsible for measuring dimensions and inspecting work pieces.
- Carry out machining activities to make Aerospace Assembly.
- Interact with inspectors, process planner, design, and tool planner.

Elements and Performance Criteria

Perform reaming operation

To be competent, the user/individual on the job must be able to:

- PC1.** Read the process sheet & drawing
- PC2.** Check and inspect the parts supplied for physical / dimensional variation.
- PC3.** Check the documentation of all parts its identification with respect to documents.
- PC4.** Check the internal diameter of bush/part by tool mentioned in technology.
- PC5.** Select proper tools and reamers as mentioned in technology.
- PC6.** Check the machinability of reamers i.e. like sharpness, size etc.
- PC7.** Select the proper coolant for reaming operation as mentioned in the technology.
- PC8.** Set the speed of rotation and feed rate for reaming operation as mentioned in the technology.
- PC9.** Select the proper sequence of reamers as per dimension, tolerance and surface finish mentioned in the technology.
- PC10.** Reaming of holes as per sequence of reamers selected.
- PC11.** Apply coolant in between the reaming operation as mentioned in the technology.
- PC12.** After reaming holes, clean the reamers and reamed holes with cotton napkin as mentioned in technology.
- PC13.** Check the final dimension, surface finish according to drawing and technology.
- PC14.** Clean the tool and gauges with the help of cotton napkin as mentioned in the technology.
- PC15.** Document the dimension.
- PC16.** Apply the non-corrosive chemical as per technology.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Understanding of reaming as a precision machining process and its role in achieving dimensional accuracy.
- KU2.** Knowledge of types of reamers (hand, machine, adjustable, etc.) and their specific applications.
- KU3.** Understanding of how to inspect internal diameters before and after reaming using appropriate gauges.
- KU4.** Knowledge of selecting proper coolants for reaming operations to ensure tool life and surface finish.
- KU5.** Understanding of setting correct spindle speed and feed rate for different materials during reaming.
- KU6.** Ability to sequence multiple reamers based on hole size, tolerance, and finish requirements.
- KU7.** Knowledge of surface finish standards related to reamed holes in aerospace assemblies.
- KU8.** Understanding of post-reaming cleaning practices and chemical applications to prevent corrosion.
- KU9.** Awareness of safety precautions specific to high-precision reaming operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Accurately read and follow reaming-related instructions in process sheets and drawings.
- GS2.** Handle and operate reaming tools and equipment with precision.
- GS3.** Set and adjust machine parameters like speed and feed for reaming operations.
- GS4.** Use appropriate gauges and instruments to measure internal diameters and surface finishes.
- GS5.** Apply coolants and cleaning methods as specified in the technology.
- GS6.** Maintain cleanliness of tools, gauges, and work areas to prevent contamination.
- GS7.** Record and document dimensional data after reaming for quality assurance.
- GS8.** Follow safety protocols related to high-precision machining operations.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform reaming operation</i>	40	60	-	-
PC1. Read the process sheet & drawing	3	3	-	-
PC2. Check and inspect the parts supplied for physical / dimensional variation.	3	3	-	-
PC3. Check the documentation of all parts its identification with respect to documents.	3	3	-	-
PC4. Check the internal diameter of bush/part by tool mentioned in technology.	3	3	-	-
PC5. Select proper tools and reamers as mentioned in technology.	3	4	-	-
PC6. Check the machinability of reamers i.e. like sharpness, size etc.	3	4	-	-
PC7. Select the proper coolant for reaming operation as mentioned in the technology.	3	4	-	-
PC8. Set the speed of rotation and feed rate for reaming operation as mentioned in the technology.	3	4	-	-
PC9. Select the proper sequence of reamers as per dimension, tolerance and surface finish mentioned in the technology.	2	4	-	-
PC10. Reaming of holes as per sequence of reamers selected.	2	4	-	-
PC11. Apply coolant in between the reaming operation as mentioned in the technology.	2	4	-	-
PC12. After reaming holes, clean the reamers and reamed holes with cotton napkin as mentioned in technology.	2	4	-	-
PC13. Check the final dimension, surface finish according to drawing and technology.	2	4	-	-
PC14. Clean the tool and gauges with the help of cotton napkin as mentioned in the technology.	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Document the dimension.	2	4	-	-
PC16. Apply the non-corrosive chemical as per technology.	2	4	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1619
NOS Name	Understand the Reaming Operation
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AAS/N1620: Understand the Canopy Matching

Description

This module/ unit is about Canopy matching ensuring the precise alignment and fitting of the canopy to the aircraft structure. The technician is responsible for verifying the clearance and fit between the canopy and the surrounding components. This process includes marking, trimming, and adjusting the canopy using specialized tools like speed guns to ensure proper fitment. The goal is to ensure that the canopy is securely installed, providing both functional and safety integrity, with clearances suitable for sealing hose installation and other critical operations.

Scope

The scope covers the following :

- Responsible for conducting drilling activities such as drilling, countersinking, counter boring, and reaming.
- Responsible for measuring dimensions and inspecting work pieces.
- Perform bolting, sealing, and other machining operations to make Aerospace Assembly.
- Operate tools and equipment such as speed gun, Vernier caliper, filler gauge, taper gauge, hydraulic trolley, and crane.
- Read and fill technical charts as per process requirements.
- Interact with inspectors, process planners, design, and tool planners.

Elements and Performance Criteria

To perform Canopy Matching

To be competent, the user/individual on the job must be able to:

- PC1.** Read the process sheet & drawing
- PC2.** Check and inspect the parts supplied for physical / dimensional variation
- PC3.** Check the documentation of all parts its identification with respect to documents.
- PC4.** Prepare Cylinder to be installed, check its certificate and install with bolting using proper tool and bolts.
- PC5.** Check whether the component having proper edge distance and other dimensions prior to carrying out operation.
- PC6.** Install hydraulics pipe connection to cylinder IN and OUT of hydraulic trolley.
- PC7.** Operate hydraulic trolley and ensure proper function of cylinder.
- PC8.** Prepare Suspension unit to be installed and check its certification. Remove hatch covers with screw gun and place hatch cover and screws in proper place.
- PC9.** Install cylinder bolting with suspension unit.
- PC10.** centralize the suspension unit and install shim according to it.
- PC11.** Drilled the holes to shim and make countersink on it and install shim with sealant, Riveting is to be carried out with proper gun.
- PC12.** Offer to inspect the working of suspension unit by operating trolley.

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- PC13.** Prepare The Canopy to be used for matching and check its certification. Check marks on bubble and apply protective casing on it.
- PC14.** Operate hydraulic trolley and check the touching area of canopy and mark on it.
- PC15.** Trimmed touching area of canopy with speed gun and ensure proper clearance. Proper gap is to be maintained for sealing hose installation.
- PC16.** Repeat the operation until canopy fully matched and proper clearance done.
- PC17.** After final matching refer to inspection and proper clearance chart is to be filled and documented.
- PC18.** Install rollers according to technology with proper clearance and drilled the holes and bolted.
- PC19.** Offer to inspect the working of rollers by operating trolley.
- PC20.** Install Dowel Pins with liquid shim i.e. ZP-3. Holes are drilled and dowel pins to be bolted.
- PC21.** Offer to inspect the working of dowel pins by operating trolley.
- PC22.** Install Canopy Supports with help of drilling fixture, drilled the holes ensure proper clearance.
- PC23.** Install Proposer with help of template, drilled the holes and ensure proper clearance.
- PC24.** Final bolting of Canopy supports and pyro pusher has to be done with sealant.
- PC25.** Offer to inspect the working of canopy support and pyro pusher by operating trolley.
- PC26.** Install manual opening handle and manual mechanism according to technology.
- PC27.** Offer to inspect the working of manual opening mechanism by operating manual opening handle.
- PC28.** Install hook to suspension unit with help of filler gauge and proper gap to be maintained.
- PC29.** Offer to inspect the gap of hook and reading should be documented.
- PC30.** Remove the canopy and fixed the canopy on sealing hose installation fixture.
- PC31.** Remove paint and other deburrs on sealing hose channel.
- PC32.** Apply protective cloth with help of glue in sealing hose channel.
- PC33.** Clean the sealing hose according to technology and apply glue to install sealing hose in channel.
- PC34.** Fixture has to install to hold the sealing hose tightly apply pressure in sealing hose to pressurize it.
- PC35.** Offer to inspect the final installation of sealing hose.
- PC36.** Finally install all canopy covers with help of screw and inspect proper safety of screw
- PC37.** Apply non corrosive paste on trimming surface and get primer on that.
- PC38.** Mark red marking on moving part.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of process sheets, assembly drawings, and how to interpret technical instructions.
- KU2.** Knowledge of dimensional inspection techniques and how to verify physical properties of parts.
- KU3.** Familiarity with part documentation, certification requirements, and part identification.

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- KU4.** Understanding of bolting operations and selection of appropriate tools and torque requirements.
- KU5.** Knowledge of hydraulic systems, including cylinder installation and hydraulic trolley operations.
- KU6.** Understanding of suspension unit alignment, shimming techniques, and the purpose of centralization.
- KU7.** Knowledge of canopy matching procedures, clearance setting, trimming, and sealing requirements.
- KU8.** Understanding of sealing hose installation processes, glue application, and pressure fixture use.
- KU9.** Familiarity with installing mechanical components like rollers, dowel pins, canopy supports, proposers, and hooks with correct tools and gauges.
- KU10.** Knowledge of protective treatments like applying primer, non-corrosive pastes, and protective cloths.
- KU11.** Understanding of inspection protocols for all installed components and documentation of results.
- KU12.** Knowledge of safety procedures while using tools like screw guns, speed guns, filler gauges, and cranes.
- KU13.** Awareness of technical chart filling and compliance with assembly technology guidelines.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Efficiently perform bolting, hydraulic pipe connection, and suspension unit installation tasks according to procedure.
- GS2.** Set up and operate hydraulic trolleys, ensuring proper function of cylinders and suspension units.
- GS3.** Maintain accurate documentation of assembly steps, including certification, clearance charts, and inspection results.
- GS4.** Safely operate mechanical tools such as screw guns and rivet guns while maintaining component integrity.
- GS5.** Perform matching and trimming operations for canopies with attention to clearance and safety specifications.
- GS6.** Apply sealants, adhesives, and protective coverings with precision during assembly.
- GS7.** Perform detailed inspections and corrections (e.g., adjusting gaps, re-sealing) as needed based on operational requirements.
- GS8.** Collaborate with the inspection team to ensure all assemblies meet quality and safety standards.
- GS9.** Adhere to safety protocols when using heavy machinery and handling materials.
- GS10.** Use technical charts and reference materials effectively to complete and document tasks according to the technology sheet.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>To perform Canopy Matching</i>	38	62	-	-
PC1. Read the process sheet & drawing	1	1	-	-
PC2. Check and inspect the parts supplied for physical / dimensional variation	1	1	-	-
PC3. Check the documentation of all parts its identification with respect to documents.	1	1	-	-
PC4. Prepare Cylinder to be installed, check its certificate and install with bolting using proper tool and bolts.	1	1	-	-
PC5. Check whether the component having proper edge distance and other dimensions prior to carrying out operation.	1	1	-	-
PC6. Install hydraulics pipe connection to cylinder IN and OUT of hydraulic trolley.	1	1	-	-
PC7. Operate hydraulic trolley and ensure proper function of cylinder.	1	1	-	-
PC8. Prepare Suspension unit to be installed and check its certification. Remove hatch covers with screw gun and place hatch cover and screws in proper place.	1	1	-	-
PC9. Install cylinder bolting with suspension unit.	1	2	-	-
PC10. centralize the suspension unit and install shim according to it.	1	2	-	-
PC11. Drilled the holes to shim and make countersink on it and install shim with sealant, Riveting is to be carried out with proper gun.	1	2	-	-
PC12. Offer to inspect the working of suspension unit by operating trolley.	1	2	-	-
PC13. Prepare The Canopy to be used for matching and check its certification. Check marks on bubble and apply protective casing on it.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Operate hydraulic trolley and check the touching area of canopy and mark on it.	1	2	-	-
PC15. Trimmed touching area of canopy with speed gun and ensure proper clearance. Proper gap is to be maintained for sealing hose installation.	1	2	-	-
PC16. Repeat the operation until canopy fully matched and proper clearance done.	1	2	-	-
PC17. After final matching refer to inspection and proper clearance chart is to be filled and documented.	1	2	-	-
PC18. Install rollers according to technology with proper clearance and drilled the holes and bolted.	1	2	-	-
PC19. Offer to inspect the working of rollers by operating trolley.	1	2	-	-
PC20. Install Dowel Pins with liquid shim i.e. ZP-3. Holes are drilled and dowel pins to be bolted.	1	2	-	-
PC21. Offer to inspect the working of dowel pins by operating trolley.	1	2	-	-
PC22. Install Canopy Supports with help of drilling fixture, drilled the holes ensure proper clearance.	1	2	-	-
PC23. Install Proposer with help of template, drilled the holes and ensure proper clearance.	1	2	-	-
PC24. Final bolting of Canopy supports and pyro pusher has to be done with sealant.	1	2	-	-
PC25. Offer to inspect the working of canopy support and pyro pusher by operating trolley.	1	2	-	-
PC26. Install manual opening handle and manual mechanism according to technology.	1	2	-	-
PC27. Offer to inspect the working of manual opening mechanism by operating manual opening handle.	1	2	-	-
PC28. Install hook to suspension unit with help of filler gauge and proper gap to be maintained.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. Offer to inspect the gap of hook and reading should be documented.	1	2	-	-
PC30. Remove the canopy and fixed the canopy on sealing hose installation fixture.	1	2	-	-
PC31. Remove paint and other deburrs on sealing hose channel.	1	1	-	-
PC32. Apply protective cloth with help of glue in sealing hose channel.	1	1	-	-
PC33. Clean the sealing hose according to technology and apply glue to install sealing hose in channel.	1	1	-	-
PC34. Fixture has to install to hold the sealing hose tightly apply pressure in sealing hose to pressurize it.	1	1	-	-
PC35. Offer to inspect the final installation of sealing hose.	1	1	-	-
PC36. Finally install all canopy covers with help of screw and inspect proper safety of screw	1	1	-	-
PC37. Apply non corrosive paste on trimming surface and get primer on that.	1	1	-	-
PC38. Mark red marking on moving part.	1	3	-	-
NOS Total	38	62	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1620
NOS Name	Understand the Canopy Matching
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

AAS/N1621: Perform RMA installation of the Aircraft

Description

The RMA Installation Technician is responsible for unpacking, inspecting, and verifying the certification of Returned Material Authorization (RMA) units before installing them in aircraft. The role involves performing manual checks, ensuring proper coupling and tolerance, maintaining documentation, and interacting with various departments to ensure quality and compliance with aerospace standards.

Scope

The scope covers the following :

- Responsible for unpacking Returned Material Authorization (RMA) items.
- Check and verify certificates of the RMAs to ensure compliance with required standards.
- Install RMAs in the aircraft as per specified procedures and guidelines.
- Ensure proper documentation of RMA verification and installation.
- Interact with inspectors, process planners, design, and tool planners for smooth execution of tasks.

Elements and Performance Criteria

To Perform RMA installations of the aircraft

To be competent, the user/individual on the job must be able to:

- PC1.** Read the process sheet & drawing
- PC2.** Checking Certificates to make sure if the unit is fit to be installed.
- PC3.** Perform manual checks on RMA to make sure its proper working.
- PC4.** Installation of RMA in the aircraft.
- PC5.** Coupling of the parts with other parts maintaining proper tolerance.
- PC6.** Cleaning of area after installation.
- PC7.** Self certification for proper installation of the unit.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of the Returned Material Authorization (RMA) process, including the unpacking and handling procedures.
- KU2.** Understanding of certification standards and procedures for verifying if an RMA unit is fit for installation.
- KU3.** Familiarity with manual testing techniques to ensure proper functioning of RMAs before installation.
- KU4.** Knowledge of installation procedures and guidelines for RMAs within the aircraft structure.
- KU5.** Understanding of tolerance requirements and methods for coupling parts during installation.
- KU6.** Knowledge of cleaning and decontamination processes following the installation of RMAs.

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- KU7.** Awareness of self-certification standards and procedures for verifying the successful installation of units.
- KU8.** Knowledge of documentation practices for recording RMA verification, installation, and inspection.
- KU9.** Familiarity with tools and equipment needed for RMA handling, installation, and testing.
- KU10.** Understanding of safety and quality control standards during the handling and installation of RMAs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret process sheets, drawings, and RMA documentation accurately.
- GS2.** Communicate effectively with inspectors, process planners, and engineers regarding RMA status and installation.
- GS3.** Perform manual inspections and functional checks with attention to detail.
- GS4.** Accurately fill out and maintain records related to RMA verification, installation, and self-certification.
- GS5.** Handle RMAs and tools safely and efficiently during unpacking and installation.
- GS6.** Identify and resolve basic issues related to RMA fit, function, or documentation.
- GS7.** Work independently while following standard operating procedures and quality standards.
- GS8.** Collaborate with cross-functional teams to ensure smooth workflow and compliance.
- GS9.** Maintain cleanliness and safety of the workspace post-installation.
- GS10.** Adhere to timelines and prioritize tasks to ensure efficient installation and inspection.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>To Perform RMA installations of the aircraft</i>	40	60	-	-
PC1. Read the process sheet & drawing	6	6	-	-
PC2. Checking Certificates to make sure if the unit is fit to be installed.	6	9	-	-
PC3. Perform manual checks on RMA to make sure its proper working.	6	9	-	-
PC4. Installation of RMA in the aircraft.	6	9	-	-
PC5. Coupling of the parts with other parts maintaining proper tolerance.	6	9	-	-
PC6. Cleaning of area after installation.	5	9	-	-
PC7. Self certification for proper installation of the unit.	5	9	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1621
NOS Name	Perform RMA installation of the Aircraft
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

AAS/N1622: Mixing of Sealants and Application of Sealants

Description

The technician will be responsible for mixing and applying various sealing compounds to aerospace panels, ensuring proper preparation, inspection, and curing. They will conduct sealing, riveting, and bolting operations while maintaining documentation and collaborating with other teams for seamless execution.

Scope

The scope covers the following :

- Responsible for mixing & preparation of variant of sealing compound such as spatula & brush variant of sealing compound U30 MES 5NT, VITEF 1NT, UT32 NT, U2 28 NT, P9, ZP3, VK-9.
- Responsible for conducting the sealing activities on panels.
- Responsible for measuring dimensions and inspect work pieces.
- The technician will carry out machining activities to make Aerospace Assembly.
- Interact with other technicians, supervisors, inspectors, Process planner, Design, Tool Planner etc.

Elements and Performance Criteria

Perform different operation of mixing of sealing compound

To be competent, the user/individual on the job must be able to:

- PC1.** Read & understand the tags of various sealing compound received from central lab.
- PC2.** Check & visually inspect the sealing compound received for physical variation.
- PC3.** Check the documentation /tags ,MTRs of all sealing compound, chemicals its identification.
- PC4.** Marking on the bins /cans of sealing compound according to tags, MTRs.
- PC5.** Check whether the components are being properly mixed & cured by carrying out sample mixing of sealing compound.
- PC6.** Select proper tool as mentioned in technology.
- PC7.** Check the documentation /tags ,MTRs of all sealing compound, chemicals for expiry dates.
- PC8.** Check the requisition slip for format, AC no, task no, qty, sign of supervisor, inspector.
- PC9.** Mix the sealing compound & other variants homogeneously as per the technology.
- PC10.** Check for quality.
- PC11.** Timely entries & documentation of sealing slips /requisition.
- PC12.** Maintaining the documentation /requisition slips.
- PC13.** Continuous check on the temperature & humidity of sealing room

Perform different Sealing compound applications

To be competent, the user/individual on the job must be able to:

- PC14.** Read the process information sheet & drawing and operations.
- PC15.** Check and inspect the parts supplied for physical / dimensional variation.
- PC16.** Check the documentation of all parts and its identification with respect to documents.
- PC17.** Marking on the part according to drawing / technology/ reference tool.

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- PC18.** Check whether the component having proper cleaning for sealing operation.
- PC19.** Select the proper tool as mentioned in the technology.
- PC20.** Select the parts for sealing as mentioned in the technology.
- PC21.** Apply Sub-layer P-9b or P-11 on the parts with hair brush for sealing as mentioned in technology.
- PC22.** Wait for 15 to 30 minutes for soaking of Sub-layer as mentioned in technology.
- PC23.** Now apply sealing compound with spatula on the parts where Sub-layer is applied.
- PC24.** Now parts are tech. bolted and rivets or bolt are inserted on holes.
- PC25.** Do the riveting or bolting as mentioned in technology.
- PC26.** Measure the bucking head and bucking length by Rivet Gauge or safety threads on nuts.
- PC27.** If there is no safety threads on nuts then install new long bolt and nuts.
- PC28.** If riveting is improper then rivets it and install new rivet and do the riveting.
- PC29.** Document the dimension.
- PC30.** Apply the non-corrosive chemical (Primer) as mentioned in the technology.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand types and variants of sealing compounds (e.g., U30 MES 5NT, VITEF 1NT, UT32 NT, U2 28 NT, P9,ZP3, VK-9) and their specific applications.
- KU2.** Knowledge of reading and interpreting tags, Material Test Reports (MTRs), and documentation related to sealing compounds.
- KU3.** Understand proper storage, handling, expiry validation, and identification of sealing compounds and chemicals.
- KU4.** Know the correct procedure for mixing and curing sealing compounds as per technology instructions.
- KU5.** Knowledge of sealing tools and application methods like spatula and brush techniques.
- KU6.** Understand the requisition and documentation process, including correct format and verification by supervisor/inspector.
- KU7.** Knowledge of quality inspection methods for mixed compounds and applied sealant.
- KU8.** Awareness of environmental conditions like temperature and humidity control in the sealing room.
- KU9.** Ability to read and interpret process information sheets and technical drawings.
- KU10.** Understand cleaning requirements and surface preparation before sealing.
- KU11.** Knowledge of the sequence of sealing application: sub-layer coating, soaking time, compound application, and riveting/bolting process.
- KU12.** Know the correct use of gauges like Rivet Gauge and safety checks for bolts and nuts.
- KU13.** Understand rework procedures for improper sealing or riveting and proper documentation practices.
- KU14.** Knowledge of applying non-corrosive chemicals or primers as per the process requirements.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Accurately read and interpret process information sheets, technical drawings, and sealing compound documentation.
- GS2.** Communicate effectively with supervisors, inspectors, and team members during sealing activities and inspections.
- GS3.** Maintain accurate and timely documentation of sealing compound requisitions, mixes, and applications, ensuring proper record-keeping.
- GS4.** Demonstrate attention to detail while checking sealing compound quality and ensuring compliance with technological specifications.
- GS5.** Ensure correct handling and storage of sealing compounds, with attention to expiry dates and proper identification.
- GS6.** Utilize critical thinking skills to identify and resolve any issues with sealing compound mixing, curing, or application.
- GS7.** Operate tools such as spatulas and brushes for sealing applications with precision and according to the technology.
- GS8.** Apply safety measures when working with chemicals, sealing compounds, and sealing tools.
- GS9.** Monitor and control environmental conditions (e.g., temperature and humidity) to maintain the quality of sealing compounds.
- GS10.** Collaborate effectively with process planners, design teams, and tool planners to ensure successful sealing operations and aerospace assembly processes.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform different operation of mixing of sealing compound</i>	13	32	-	-
PC1. Read & understand the tags of various sealing compound received from central lab.	1	2	-	-
PC2. Check & visually inspect the sealing compound received for physical variation.	1	3	-	-
PC3. Check the documentation /tags ,MTRs of all sealing compound, chemicals its identification.	1	2	-	-
PC4. Marking on the bins /cans of sealing compound according to tags, MTRs.	1	3	-	-
PC5. Check whether the components are being properly mixed & cured by carrying out sample mixing of sealing compound.	1	2	-	-
PC6. Select proper tool as mentioned in technology.	1	3	-	-
PC7. Check the documentation /tags ,MTRs of all sealing compound, chemicals for expiry dates.	1	2	-	-
PC8. Check the requisition slip for format, AC no, task no, qty, sign of supervisor, inspector.	1	3	-	-
PC9. Mix the sealing compound & other variants homogeneously as per the technology.	1	2	-	-
PC10. Check for quality.	1	3	-	-
PC11. Timely entries & documentation of sealing slips /requisition.	1	2	-	-
PC12. Maintaining the documentation /requisition slips.	1	3	-	-
PC13. Continuous check on the temperature & humidity of sealing room	1	2	-	-
<i>Perform different Sealing compound applications</i>	17	38	-	-
PC14. Read the process information sheet & drawing and operations.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Check and inspect the parts supplied for physical / dimensional variation.	1	3	-	-
PC16. Check the documentation of all parts and its identification with respect to documents.	1	2	-	-
PC17. Marking on the part according to drawing / technology/ reference tool.	1	3	-	-
PC18. Check whether the component having proper cleaning for sealing operation.	1	3	-	-
PC19. Select the proper tool as mentioned in the technology.	1	3	-	-
PC20. Select the parts for sealing as mentioned in the technology.	1	2	-	-
PC21. Apply Sub-layer P-9b or P-11 on the parts with hair brush for sealing as mentioned in technology.	1	2	-	-
PC22. Wait for 15 to 30 minutes for soaking of Sub-layer as mentioned in technology.	1	2	-	-
PC23. Now apply sealing compound with spatula on the parts where Sub-layer in applied.	1	2	-	-
PC24. Now parts are tech. bolted and rivets or bolt are inserted on holes.	1	2	-	-
PC25. Do the riveting or bolting as mentioned in technology.	1	2	-	-
PC26. Measure the bucking head and bucking length by Rivet Gauge or safety threads on nuts.	1	2	-	-
PC27. If there is no safety threads on nuts then install new long bolt and nuts.	1	2	-	-
PC28. If riveting is improper then rivets it and install new rivet and do the riveting.	1	2	-	-
PC29. Document the dimension.	1	2	-	-
PC30. Apply the non-corrosive chemical (Primer) as mentioned in the technology.	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AAS/N1622
NOS Name	Mixing of Sealants and Application of Sealants
Sector	Aerospace and Aviation
Sub-Sector	
Occupation	Aircraft Assembly, Testing and Overhauling
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AAS/N1616.Understand the Assembly Drawings.	40	61	-	-	101	10
AAS/N1617.Understanding the Different Drilling Operations.	40	60	-	-	100	10
AAS/N1618.Understanding of Different types of Riveting Operations	40	60	-	-	100	15
AAS/N1619.Understand the Reaming Operation	40	60	-	-	100	15
AAS/N1620.Understand the Canopy Matching	38	62	-	-	100	15
AAS/N1621.Perform RMA installation of the Aircraft	40	60	-	-	100	10
AAS/N1622.Mixing of Sealants and Application of Sealants	30	70	-	-	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	288	463	-	-	751	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
MS	Material schedule
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
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Qualification Pack

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Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
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Qualification Pack

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